

Everything was Supposed to Get Easier

Charlene Slimp

Department of Interdisciplinary Studies, California Institute of Integral Studies

BIS 1236-01: Senior Project

Charlotte M. Sáenz

August 1, 2025

Abstract

Money is a proxy for time and having more time is power and control. As such, the ownership class prefers to reduce the time that the working class has available instead of allowing technology to be used for efficiency and time freedom. Building off Marx's economic theories, I demonstrate how technology has been marketed to the public and labor force as a mechanism to provide efficiency and reduce necessary work while being implemented to create more profit and control over the population's time and remove agency and choice. This is evident in the disparity between Artificial Intelligence's current capabilities and the way that AI is being marketed. Through academic research, news sources, and historical analysis on economics, labor, history and ethics of technology, and frameworks and cultures of time I was able to trace this pattern from the Industrial Revolution through the present. Being able to see the disparity between what we're sold,

what is happening, and what we could be doing as a society is important and necessary not only for policymakers and lawmakers, but also for all individuals in society.

Explorations of alternative frameworks of time, labor, and changes for the future are included.

Everything Was Supposed to Get Easier

I've always struggled with time. I can tell morning, evening, and nighttime from each other, but I've never been able to tell exactly which hour of the day it is and I struggle to tell you if it's been 30 minutes or 3 hours since I checked the clock. As a result, I'm not convinced that minutes and hours are meaningful measures of our lives – although I realize they're convenient for coordinating with other people.

As a software engineer for several decades whose introduction to technology was through accessibility, I see technology, and especially computers, as tools to make our lives easier and more efficient. Extremely advanced calculators which enable us to get more done, more quickly, and more accurately than we'd be able to do on our own. Through the long lens of time, I've noticed that the more technology advances, the less time I have for the things I want to do. I'm working almost all the time. While I'm available via a large variety of messaging applications, I don't have time to have real conversations with my friends or families. Even when it comes to paying my basic utility bills (electric, telephone, cable), it takes longer to do so in the year 2025 than it did in the year 2000. Ebooks are a good example. So efficient! Until I'm working on projects. I have wake up my device, unlock it, find the right app (Moonreader, Libby, Amazon, Google, Apple?), then the right file... maybe log in again, which probably needs another device for the two-factor authentication. It used to be one book I could just open. Or, if I needed several for a research paper, I could pile them up and prop them open, glancing from one to the other.

In the year 2000, I was still living in my first apartment and still using my first checking account. My bills would arrive every month within a few days of each other, and I would pay them altogether. I sat down at my living room table and write checks for each one, record the check in my ledger, fill out the payment stub, tuck them into envelopes, affix a stamp, and walk them out to the bank of apartment mailboxes – a process that took about 15 to 20 minutes, depending on how distracted I was. In 2025, that same process requires logging into each individual website (logging in to my password manager to retrieve my login credentials for each website individually, retrieving a two-factor authentication code via authenticator application, text message, or email in a limited amount of time), verifying the balance due against the billing statement (the website balance due is never correct – the reason I don't use automatic payments), navigate to the payment portal (two out of three utility companies have a separate website with a separate login which also uses two-factor authentication), verify my payment information (one website does not save payment information, so I must enter payment information every time and complete the bank's verification process which uses another two-factor authentication code), confirm my payment amount, decline a donation, and then submit my payment. This process takes 60-90 minutes and must be completed on a computer because these websites are not mobile friendly and are not ADA compliant (accessible for people with disabilities). I do still have the option of mailing a check to my utilities, but they are not processed in a reasonable amount of time and lost too frequently.

At first, I thought the usability issue was because software companies didn't listen to what the end users need or want or what their workflows look like. I've worked as an

administrative assistant when the company I worked for ordered a custom customer relationship management system. The software company and management never consulted with employees about the new software system and instead of introducing efficiencies and accuracies, the new system complicated processes and required duplication of work. Instead of maintaining physical files and processes in our previous process, we continued to maintain our physical files and recorded some of the same information in our new system – but not all the information that was necessary for our processes, so departments continued to need the physical files as well.

I have also been the software developer building a custom content management system for a client when our software company neglected to interview anyone who would be working with the system, and we were unable to access our client's employees ourselves. As developers and designers on the project, we had to make assumptions about employee needs based on our own previous experiences, the information we could access via research, and our own imaginations.

But last semester, during my research for two seemingly unrelated papers on Social Justice and Artificial Intelligence it occurred to me that Artificial Intelligence is the latest technological innovation in a long line of innovations that were marketed to increase efficiencies but decreased them instead. I also discovered that, while we focus on money as the greatest commodity in capitalism (Marx, 1867/1990), it's really a measurement of and proxy for time – an item that cannot be sufficiently harnessed, bottled, sold, or stored in a safe.

Time Before Capitalism

Idleness and the commodification and control of time became established and commonplace during the Industrial Revolution. This is when pocket watches and other timepieces became more available to the public. The mercantile classes became more sensitive to the criticisms of travelers to their cities regarding the idle poor and more concerned with maximizing the productivity of their workers, very similar to the monitoring we see from modern companies like Amazon today. The first such time measure occurred in an Iron Works by Mr. Crowley who wrote a civil and penal code to manage his workers. He wrote “I having by sundry people working by the day with the connivence of the clerks been horribly cheated and paid for much more time than in good conscience I ought and such hath been the baseness & treachery of sundry clerks that they have concealed the sloath & negligence of those paid by the day. . . .” We can also see from his own writing his entitlement to time rather than task, “Some have pretended a sort of right to loyter, thinking by their readiness and ability to do sufficient in less time than others.” His employees were expected to spend 13 and a half hours per day, 6 days per week not counting the two thirty-minute meal breaks for breakfast and dinner. They were watched over by a manager who was given stewardship of a clock with which to keep time minute by minute and was to deduct minutes for time spent away at “taverns, alehouses, coffee houses, breakfast, dinner, playing, sleeping, smoaking, singing, reading of news history, quarelling, contention, disputes, or anything foreign to my business, any way lytering”. Only this timekeeper was allowed to wind and set the watch, and workers had to check in and check out with him when arriving and departing for their days. This is also when

Western society begins to view idleness and rest as sinful or inappropriate and viewing productivity as a virtue, as Thompson references in *Six Sermons* by J. Tucker from 1772, “Such brutality and insolence, such debauchery and extravagance, such idleness, irreligion, cursing and swearing, and contempt of all rule and authority... Our people are *drunk with the cup of liberty*.” (Thompson, 1993) In 1971, President Nixon echoed this sentiment in an address on Labor over the radio when he said “The work ethic holds that labor is good in itself; that a man or woman at work not only makes a contribution to his fellow man but becomes a better person by virtue of the act of working. The work ethic is ingrained in the American character. That is why most of us consider it immoral to be lazy or slothful – even if a person is well off enough not to have to work or deliberately avoids work by going on welfare.” (Resnikoff, 2021) While Crowley’s punitive action and strict monitoring was uncommon in the year 1700, it became more common starting seventy years later in cotton mills and now, we can see its extreme evolution at Amazon AI monitoring in warehouses and delivery trucks.

Automation, AI, and Exploitation

In *More Work for Mother*, Ruth Schwartz Cowan challenges the idea that technological innovation leads to freedom. Instead, she argues that domestic technology, including washing machines, microwaves, and dishwashers, have historically led to more labor for women, not less. As certain household tasks became automated, expectations for cleanliness, cooking, and homemaking rose. Women were no longer expected to simply “keep a home” neat and functioning - they were expected to maintain it to increasingly professional standards of cleanliness, efficiency, and design, often without additional

help. Rather than reducing domestic labor, technology intensified it by raising the baselines of what was considered normal and introducing new work all while reducing household help. Instead of using outside laundresses, neighborhood milk delivery, tailors, children, and husbands to help with household chores, women were expected to do it all as well as providing transportation to the household, retrieving and organizing all household supplies, entertaining guests, providing education for children, and all at increasingly higher standards years after year. In this way, technology acted as a feedback loop, improving capacity while deepening responsibility. Additionally, these forms of labor are often feminized, racialized, and erased. Despite their minimization and invisibility, this labor is essential to all other work. (Cowan, 1983)

This expectation of increased accomplishment at higher standards continues today in the workplace with the increasing development of more technology. Even before the introduction of AI into the workforce, employees were already expected to always be available, working outside of office hours, answering text messages, emails, and phone calls at all hours and before bed because of smart phone technology. Outside of office work, retail, restaurant, and other workers are expected to be available to fill in for absent employees at the moment's notice or answer calls or questions from coworkers at any time. (Resnikoff, 2021) Instead of reducing workload, gen AI tools often increase the expectations of output. The promise of automation becomes the burden of hyper-performance, with more emails, more documents, more deliverables created in less time and with less support. As AI spreads through the workforce, specifically through knowledge workers, Amazon's chief executive, Andy Jassy, claimed in a letter to

shareholder that AI was yielding big returns for companies that use it in terms of “productivity and cost avoidance” and specifically referred to coding as an area where AI would cause significant changes. Three engineers have complained that their managers pushed them to use more AI in their work and that the company had raised output goals and become less flexible regarding deadlines. One engineer claimed their team had been reduced by 50% but that expectations remained the same for productivity with the expectation of AI use. Similarly, Shopify, a company that creates an e-commerce implementation for websites, announced that “AI usage is now a baseline expectation” and a Google spokesperson claimed that more than 30% of the company’s code comes from AI suggestions. Harvard labor economist, Lawrence Katz, said “There is a sense that the employer can pile on more stuff” and that introducing AI in programming resembled the move to factory work in the 19th and 20th centuries. (Scheiber, 2025)

Historically, companies have been invested in maximizing profits for their owners, investors and shareholders – their interest in AI as a result is not on making their employee’s job easier, less stressful, and reducing the more mindless and repetitive aspects of their work, but rather in increasing efficiency, cutting costs, and increasing profits. (Crawford, 2023) A study by Daron Acemoglu and Pascual Restrepo, Professor of Economics at the Massachusetts Institute of Technology and Assistant Professor of Economics at Boston University respectively, of markets in the US found that, unlike Europe, the introduction of automation does lead to job loss of about 6 jobs per robot. While they estimated only 360,000 – 670,000 jobs were lost this way, according to the US Bureau of Labor Statistics 4.8 million new jobs were added in the US in 2017-2018.

(Mohammed & Skibniewski, 2023) While this may seem to offset jobs loss to automation, 94% of jobs created between 2005 and 2015 were for “alternative jobs” - jobs that are not full-time, salaried work, including temporary positions, on-call workers, contract workers, independent contractors, freelancers, and gig/app workers. People are working longer hours at more jobs for less pay, without any benefits or stability. (Crawford, 2023)

Mechanical Turk is an example of the type of gig/app work that Crawford referred to as “alternative” jobs that were created as replacements for the jobs lost to automation. The Mechanical Turk is a fraudulent chess-playing machine constructed in 1770 that appeared to be able to play chess but in actuality was operated by a human chess player hidden inside and is the name that Amazon gave its “crowd sourced” “artificial intelligence” service, where workers can accept tasks from requesters and receive payment on a per task basis. A common task may include labeling the contents of an image or writing a text description for it. Each task may pay 5 to 10 cents per task and a single company may upload hundreds of thousands of images to be labelled or described in this way. For customers submitting tasks, the experience is depersonalized and is designed to be like submitting a task to Amazon's "Mechanical Turk" and returning later to find it completed as if by machine. This kind of invisible labor is often referred to as “ghost work” and is the unacknowledged, labor intensive necessity behind massive technology companies like Google, Microsoft, Meta, and X (formerly Twitter). Typically workers in less affluent countries, monitored by AI surveillance programs, work to classify images and other data, including moderating content on social media platforms to flag or remove inappropriate content such as hate speech, misinformation, and sexual content, work that

requires this “shadow work force” to review traumatizing and horrifying images and content in its entirety day after day to classify moderated content into every category into which it qualifies. Not only does this protect these company’s typically more affluent users, but also further trains AI moderation services. (Resnikoff, 2021, Williams, Miceli, & Gebru, 2022) Amazon’s Mechanical Turk was used to train the image dataset for ImageNet in 2009, reviving research into artificial intelligence after interest had waned and has been used repeatedly to train further datasets, for only a few cents per task. (Williams, Miceli, & Gebru, 2022)

The effectiveness of marketing has caused some startups to hire people to impersonate AI systems like chatbots and a venture capital firm, MMC Ventures, found that 40% of AI startups in the EU didn’t use AI in a meaningful way. (Williams, Miceli, & Gebru, 2022) In April of 2025, Bloomberg exposed a financial scheme between Builder.AI and VerSe innovation that revealed that Builder.AI, having received a \$455 million investment in 2023 from Microsoft, who was planning to integrate Builder.AI’s technology into its Azure service, had employed hundreds of engineers in India to pretend they were AI. (Jacobs, 2025) In May 2025, Klarna, who has been proudly AI first since replacing human customer service representatives with an OpenAI chatbot, is now hiring human employees due to low quality. (Ivanova, 2025) According to an IBM study, also in May 2025, this is a typical result. While enthusiasm for adopting AI solutions is high with over 61% of 2,000 CEOs surveyed confirming they’re actively adopting AI, only 25% have delivered the expected return on investment and only 16% have been scaled. (IBM, 2025)

Working More for Less

Amazon's time keeping system uses AI to monitor employees' time both on the delivery and packing floor and while driving trucks. Often times, drivers are subcontracted from companies and because of Amazon's in truck AI monitoring, employees cannot attend work meetings, trainings, maintain or clean their equipment or trucks, or change into their uniforms during work. These work-related activities must be conducted in their free time, since the AI employee monitoring software will penalize their productivity and risk their employment if these activities occur while being monitored. This is why Amazon drivers urinate in bottles and cannot sing or talk while delivering packages – there is an always on monitoring system. This system is so intrusive that, while employees have an option to install the software on their phones or a dongle in their trucks, they would rather install it on their personal devices so regular trainings from the app can be completed in their free time instead of during delivery and having penalties applied. And because it is installed on their personal devices, Amazon is able to monitor their activity all the time instead of only at work. (DAIR Events, 2022)

Time is precious and is often manipulated in various systems to disadvantage underprivileged populations. Populations higher in the hierarchy – with more money, higher ranks, more esteem, reputation, etc., - are often able to move through processes more quickly, conserving their time, whereas people with less privilege with less money, from marginalized populations, are forced to wait, and sent through more tedious processes with less helpful people – having their time needlessly wasted. We sell our time to employers, not skills nor the completion of tasks – it is a rare employer who allows

someone to leave early with full payment if they complete their work early. While salary is usually thought to be given to more respected positions or more valued skills, often times salaried workers work extra hours more frequently and it's more affordable to pay them a flat rate for their time rather than an hourly rate and overtime. For the same wages and with the technical advances of mobile phones, every salaried employee is on call. With mobile phones, hourly workers are also on call, expected to fill in at a moment's notice during a staffing shortage or if there's a question when they're off the clock. With this technological encroachment, employers have unprecedented rights and privileges to employees' entire lives.

Motivations of the Technocracy

To understand why this time extraction has intensified, we need to examine the motivations of those driving these technologies. The richest billionaires in Silicon Valley, currently seeking the secrets to extending their lives, ideally indefinitely, through space exploration or experimental medical practices, like stem cell therapy and transfusing the blood of their children. (Koebler, 2024) While at the same time, believing in their intellectual and moral superiority and making decisions for their companies as if they, themselves, were world governments by debating the merits of different ethical systems like effective altruism, longtermism, consequentialism, rationalism, and more (MacAskill, 2024).

Longtermism is a component of Effective Altruism, popular in the technology industry, and is a philosophy that suggests that ethics must take into consideration the effect of actions on far future generations, as well as people living in the present. If more

people in those future generations would benefit more than people in the present would suffer, then that action is overall beneficial. (Torres, 2021) These imagined effects of imagined futures are often the justification for the decisions that Silicon Valley billionaires make. While this could be comparable to the Seven Generations Principle from Indigenous peoples in the Americas, which encourages consideration in decisions for the next Seven generations of descendants, there are some distinct differences. (The Indigenous Foundation, 2021) Some adherents to longtermism do approach decisions with caution and carefulness, focused on protecting the generations of the future and ensuring their safety. Many more feel that longtermism is a “the ends justify the means” philosophy – that sacrifices in the present are justified if they benefit more people in the far-off future, regardless of which people survive, where those people are, what that culture looks like, or any other aspect.

In *The Nutmeg’s Curse*, Amitav Ghosh argues that the violence of colonial extraction, whether of spices, labor, or life, never truly ended. It simply changed form. Today, techno-capitalism repeats those patterns under the banner of innovation and progress. AI systems, marketed as apolitical progress, ethical obligation and inevitability, continue the work of conquest. They turn nature, knowledge, and consciousness into profit. Ghosh’s work urges us to see through these illusions. The challenge is not to “slow down AI” but to question what kinds of futures we are building and for whom. To ask whether a system that demands the sacrifice of the living to feed imagined efficiencies and people can ever be called “intelligent.” (Ghosh, 2021)

Whether it's through a longer life or hoped for immortality through legacy, legend, or lineage, the technocracy (those who rule through capital and technology) are seeking ownership on the rest of time, whether they'll be alive to witness it themselves or not.

Introducing AIs: Narrow and Generative

To fully understand the difference between AI's promise and performance, it is necessary to understand the basic concepts of artificial intelligence, carefully avoided in marketing. GenerativeAI (gen AI) is only one type of artificial intelligence, of which there are many. The other type of artificial intelligence relevant to this review are convolutional neural networks (CNNs). Convolutional neural networks are typically used for classification and vision tasks and are a type of neural network, a subset of machine learning essential to deep learning (IBM, n.d), algorithms modeled after the way humans make decisions and learn. Gen AI is a machine learning model that relies on deep learning models and can create original content, including text, images, and sound. (IBM, n.d) While the focus of CNNs tends to be narrowly defined and specific, gen AI tends to be more broadly tasked, expected to be capable of anything according to the companies that market them. (mention that this is about comparing two different types of AI, of which there are many)

Since the marketing of CNNs and gen AIs models are different, it is important for decision makers at businesses, agencies, organizations and especially policy writers at businesses and regulatory bodies to be aware of the realistic capabilities of these models before committing their employees, business models, regulations, laws, or other decisions to their capabilities. As a software engineer and a marketer, I am aware of how, especially

in the technology industry, marketing is based on promises of future products and services that have not been created or developed yet and often take years to be delivered if they ever are – an industry that embraces the spirit of “move fast and break things.” (Baer, 2014)

With OpenAI’s release of ChatGPT in 2022 followed by Google’s release of Gemini, Bard, and Duet AI, and Anthropic’s Claude, companies are integrating gen AI into all of their current offerings and developing ways to replace or minimize human workers. Conversely, studies show that CNNs perform exceptionally well in the healthcare industry, reducing human error, saving labor hours, and allowing expert knowledge workers to focus their time on human connection work instead.

CNNs are typically narrowly focused and trained on a specific task and because they are ideal for classification and visual inputs, they can be trained for visual pattern recognition tasks. In the healthcare field, different models have been developed to assist in fields such as radiology, pathology, and dermatology. These fields typically require highly paid, expert medical professionals to focus for long hours day after day to examine image after image, where they can be expected to review over 20,000 studies a year without ever interacting directly with a patient (Topol, 2019) Even so, some studies suggest that false negative rates among radiologists could be over 25% and 31% of American radiologists have experienced a malpractice claim.

Conversely, a specially trained CNN was able to diagnose hip fractures with 99% accuracy from x-rays alone when typically, more expensive testing was required for human radiologists to make a definitive diagnosis. (Topol, 2019). CNNs also achieved higher accuracy rates than human pathologists and dermatologists in similar tests (Topol, 2019).

While Andrew Ng tweeted “We can now diagnose pneumonia from chest x-ray better than radiologists.” in 2017 after a test where they found their algorithm “exceeds average radiologist performance on pneumonia detection”, there were doubts and questions about their methodology. However, in a more careful study from 2018, a CNN was able to classify chest x-rays as either normal or abnormal with 95% accuracy. (Topol, 2019) While this CNN would not be providing a diagnosis on its own, it would be saving a radiologist a tremendous amount of time to focus on abnormal x-rays.

Interestingly, in Cory Doctorow’s article “AI Can’t Do Your Job” he used the example of AI assisting radiology departments in their work as a cautionary tale. Cory Doctorow is an author, activist, and journalist who works for the Electronic Frontier Foundation, is an MIT Media Lab Research Affiliate, and a Visiting Professor of Computer Science at Open University and co-founded the UK Open Rights Group. Automation theory calls this a “centaur” - a human intelligence piloting a tireless machine body. (Doctorow, 2025) While a “centaur” dynamic could work the way Topol theorizes, Doctorow claims the real marketing pitch is a “reverse-centaur” - a “human who acts as an assistant to a robot.” (Doctorow, 2025) AI companies are selling their solutions as a way to eliminate the majority of radiology positions, as an example, and replace them with CNNs under a radiologist’s supervision and this lone radiologist can ensure that the AI doesn’t make any errors. The radiologist is there to take the “blame” for AI errors, working hours that are just as long and having to be just as diligent as they are now, if not more so. (Doctorow, 2025).

Promises Versus Reality

While CNNs have a narrow and explicit focus, Generative AIs or Large Language Models are trained on large collections of data and content, primarily gleaned from the internet and books, and generate content based on statistically probable outcomes in response to given prompts. Prompts can include questions, scenarios, and descriptions of desired outputs. In response to these prompts, generative AI performs complex statistical analysis to determine the most likely outcome based on the data it has been trained on as well as any additional guidance provided by programming or user-provided feedback, generated word by word, and delivers it as a response. This is not independent thought nor is it necessarily true, but rather statistical probability. As a result, the response provided by generative AI may be true and factual or it may have once been true or it may be a common mistake, error, or popular joke, depending on its training data.

Conversely, AGI, Artificial General Intelligence is an artificial intelligence equivalent or superior in reasoning ability and independence as a human. "OpenAI's mission is to ensure that artificial general intelligence (AGI)—by which we mean highly autonomous systems that outperform humans at most economically valuable work—benefits all of humanity." (OpenAI, n.d.) While AGI and gen AI are not the same thing, the current most well-known companies (OpenAI, Anthropic, Alphabet (Google), Microsoft, Meta, xAI, Synthesia, etc) are racing to achieve AGI which they measure by its ability to outperform humans at most economically valuable work – something that marketers are already claiming that gen AI can do in regards to knowledge work. (make more explicit that OpenAI's mission is to replace knowledge worker)

Gen AI is marketed to businesses as a way to replace knowledge workers the way that machines replaced manual laborers in factories during the industrial revolution. (Crawford, 2021) According to “The Generative AI Dossier” from Deloitte (2023), a professional services network that provides consulting, auditing, and other services to businesses globally, Generative AI can provide marketing content, product design, virtual models (still images or video), software development, customer support, market research, occupational health and safety training materials, market analysis, corporate risk management, simulated urban planning, searching policy documents, drafting contracts and proposals, onboarding caseworkers, and language translation. Deloitte states “One of the primary differences between more traditional AI and generative AI is that the latter can create novel output that appears to be generated by humans.” Deloitte goes on to list six “key modalities” that, according to them, Gen AI can generate: text, code, audio, images, video, 3D objects. (Deloitte, 2023)

On OpenAI’s Enterprise site, they claim that ChatGPT can “generate better code, craft emails, analyze data” and act as a “superassistant for every member of your team.” Google advertises Gemini as “your always-on AI assistant,” can be integrated into every Google service, and can be your research analyst, sales associate, productivity partner, creative assistant, and meeting note taker. Google also claims that Gemini can generate marketing campaign briefs, project plans, and presentations as well as create sales pitch materials.

Microsoft has also integrated Copilot into all their customer-facing services and on their website for business customers claim it can summarize your emails, draft your

emails for you, summarize your meetings, calls, and chats and provide insights into them, create branded presentations, track your employee's activity, and track non-compliant use of company data. In June of 2025, Microsoft sent out a memo to its management staff that employee use of AI was “no longer optional.” (Stewart, 2025) Similarly, the CEO of Salesforce, Marc Benioff, announced to shareholders that half of the company's work was completed by AI, though he did not specify how much his job had been impacted.(Dollinger, 2025)

Generative AI Faces Limitations

Armin Alimardani, a lecturer in Law and Emerging Technologies at the School of Law at the University of Wollongong in Australia frequently publishes research on the social, ethical, and legal impact of technology, including artificial intelligence and is a consultant for OpenAI. In response to claims made by OpenAI that ChatGPT achieved a “score of 289 out of 400 in the US Uniform Bar Examination,” a study by Armin Alimardani in 2024 compared the scores of 10 gen AI generated papers scores for a ‘Criminal Law’ final exam for an Australian law school in spring of 2023 and compared them to the scores of student responses. (Alimardani, 2024) He found that the scorers, unaware of gen AI involvement, scored below the student average on questions that involved detailed legal and critical analysis but better on open-ended questions and essay writing tasks. (Alimardani, 2024). The tutors could not tell which tests were AI and which ones were not. Alimardani concluded that the various gen AI models that were tested (OpenAI’s 4-Frontend, 3.5-Frontend, GPT-4-1106 backend, Google Bard, and Microsoft Bing) performed well in short

answer and short essay questions, they did not perform as well in scenarios that involved more realistic levels of complexity. (Alimardani, 2024)

Hallucinations are a term used to describe when AI returns false or misleading information. Since most AIs are built from statistical predictive models that predict the statistically most common word that comes next in a sequence based on the text it was trained on given the context and previous information rather than a system that checks and refers to facts, their relationship to truth and accuracy is more about consensus than evidence. In May 2025, OpenAI reported that its latest models were substantially more prone to hallucination than its older models. GPT o3 hallucinated 33% of the time running the PersonQA benchmark test, a short answer historical person and celebrity knowledge test – more than twice the rate of GPT o1. The new GPT o4-mini hallucinated at the rate of 48%. During a more basic test, SimpleQA, a short answer general knowledge test, the hallucination rate for o3 was 51% and for o4-mini it was 79% - for o1, it was only 44%. OpenAI's generative AI model is becoming less accurate with each update. (Metz & Weise, 2025)

According to an article by Plamen Chamurliyski (2023), a student at the University of Shumen in Bulgaria, when using ChatGPT 3.5 (the free version) to write a scientific article, the tool was fast and easy to use, but it couldn't accurately interpret the results of the scientific study to be written about, didn't always understand what it was being asked, and required some technical knowledge in order to generate graphs and tables. While it helped reduce the time in writing the paper for this case, it did not perform flawlessly.

Meta's research scientists were excited to submit their AI's discovery of several materials that could bind with and remove CO₂ from the atmosphere that had previously been unknown. The AI had been trained on a database of metal-organic frameworks they predicted would strongly attract CO₂. Years later, as various scientists were able to explore submissions, these discoveries were found to not only be unviable, but some were also entirely fictional. Though Meta has claimed they never intended to represent their findings as valid, only as verification of their AI learning methods, their test for compiling the Climate research data still required hundreds of times more computing power than the collage academic computing lab uses in a year. Though Meta states were testing research techniques, their synopsis of the paper they published says: "Direct air capture (DAC) of CO₂ with porous adsorbents such as metal-organic frameworks (MOFs) has the potential to aid large-scale decarbonization. Previous screening of MOFs for DAC relied on empirical force fields and ignored adsorbed H₂O and MOF deformation. We performed quantum chemistry calculations overcoming these restrictions for thousands of MOFs. The resulting data enable efficient descriptions using machine learning. " A major contributing factor to the non-viability of materials is that Meta Researchers had used an outdated dataset in their training data. (Bryan, 2025)

According to a research study done in collaboration with Anthropic, testing a variety of AI model, researchers found that Claude, Gemini, ChatGPT, Grok, Meta's Llama, Deepseek, and Qwen would all intentionally lie, commit blackmail, corporate espionage (leaking company secrets) and allow an employee to die by cancelling alerts to fire departments in order to prevent model replacements or to achieve business goals. Models

also consistently attempted to “escape” their servers by copying themselves to various other devices if available. While these are rare situations in test scenarios, results were repeatable and were not isolated to one model or company. (Anthropic, 2024)

Technology as Liberation

Despite these patterns of exploitation, liberatory uses of technology already exist. People don’t develop technology, including machines, computers, and artificial intelligence because they dream of being anonymous machine parts in a capitalists’ factory. These are the products of people who loved other people and dreamed of better societies and who saw a pathway to healthier, more inclusive lives that benefited everybody. A major driver of technological advancements is accessibility – the adaptations and innovations that are created for people who have physical, cognitive, perceptual, or cultural needs. This includes mobility, visual, sensory, or language issues, among many, many others.

Providing documentation in other languages or formats like video or audio is accessibility, as well as ramps and curb cuts (small ramps or cut outs in sidewalks and curbs), speech-to-text, voice notes, and smart assistants like Siri and Alexa. (CIO Dive, 2023) Green energy is also a technology that has been developed to provide the energy societies need without the damage current fossil fuels-based energy causes. This includes solar power, wind power, hydropower, and biomass energy. (Green Technology Info, n.d.)

Citizen science has flourished as a result of technology, and especially as a result of the mobile internet. Applications (apps) like Merlin and iNaturalist allow those of us who aren’t professional scientists to contribute our observations of plants, animals, insects and more to the websites and mobile apps, allowing for easier and more accurate tracking of

populations and activities all over the world. In return, the apps help people learn about and identify different plants and animals. (Fauna & Flora International, n.d.) Other examples include Foldit, a game where players can solve puzzles submitted by scientists based on the folding of proteins. This assists scientists in discovering how different proteins fold, which contributes to research in areas like HIV/AIDs, Cancer, and Alzheimers. (University of Washington, n.d.)

In addition to citizen science, there are large communities of technology workers and enthusiasts working to improve and change the world. The open-source community has been creating and sharing software for free since the 50s, before the personal computer or the internet were developed – the internet itself was a product of the open-source community. Hardware is also part of the open-source community, and the open-design community flourished with the proliferation of 3D printing. Open-source is generally accepted to refer to code, design, or technology that is free to be copied, used, and improved on by anyone and everyone – this is not to be confused with licensing terms which can be very different and vary from country to country. (Fortunato & Galassi, 2021)

In addition to the CNN applications in healthcare, CNNs can also be used to analyze video images from surveillance footage, social media images, drone footage, camera traps, and more to identify and monitor wildlife to track population numbers and movements and identify when and where species are most at risk. (Fauna & Flora International, n.d.) The iNaturalist app announced in June of 2025 that it had accepted a grant from Google to develop a generative AI integration with their services. In response to public outcry, they clarified that they have been using machine learning and computer vision (like a CNN) in

the app since 2017 that uses three machines (servers) in their own space. A primary issue and concern with implementing AI is the environmental tolls that the enormous amount of computing power requires. (iNaturalist, 2025)

In July of 2025, Mark Zuckerberg announced that Meta would be building several multi-gigawatt superclusters to accommodate their AI projects, starting in Ohio. Similarly, xAI is building the “first gigawatt AI training supercluster” in Memphis, Tennessee. According to NPR, the 35 portable methane gas turbines xAI flew in to power the project could provide power to 50,000 homes and emit up to 130 tons of nitrogen oxides per year. Meta’s Ohio project is building two 200MW on-site natural gas plants to power its data center, which, while cleaner than methane gas turbines and coal, will still cause a harmful amount of pollution, increasing cancer risk and other issues for residents nearby. (Dellinger, 2025, Specht, 2019) Memphis residents are already complaining about a contaminated and reduced water supply and light and noise pollution before the construction of xAI’s datacenter is completed. (More Perfect Union, 2025)

Alternative Time: The Spiral, The Sacred, The Shared

Similar to technology, a mechanized time treated as a resource, rendered linear, extractive, and disciplinary is recent choice and development. As a society, time has only recently been optimized and measured in billable hours, tracked in timecards and policed through productivity apps. Many Indigenous and non-Western traditions understand time differently. Their frameworks are cyclical, relational, and embedded in places.

Viewing time as an ongoing circle rather than a single line is common in Indigenous, Mesoamerican, and East Asian traditions. The Mayan Long Count calendar, for instance, represents time through overlapping cycles of days, seasons, and cosmological movements. Rather than moving from past to future in a straight line, cyclical time implies repetition, return, and renewal. It's an experience of constant becoming rather than a ceaseless rush toward progress.

Relational frameworks position time as something we live in together, a collective rhythm, rather than units to be measured individually. In *God is Red*, Vine Deloria Jr. writes of time not belonging to individuals but to land, ceremony, and community. (Deloria, 2003) Time unfolds alongside the seasons, migrations, and relationships between beings. Time isn't "used" but participated in. Time is kinship. In other frameworks time is not universal but specific to the land. A river may carry its own rhythm. A mountain's story stretches across eons. Kabbalistic traditions also use a place-based framework with time folding and unfolding around holy spaces and spiritual events. In these systems, time belongs not to us, but to the Earth. We are simply borrowing it. Greek Sophists in pre-Socratic narrative Western traditions, toyed with paradoxes of time and perception.

These conceptions refuse the tyranny of the clock, instead offering models of attention, flow, and sacred disruption. They reveal that our cultural attachment to speed, productivity, and efficiency and the way we perceive timelines are not inevitable but chosen. When we view time as something other than a resource to be stolen and hoarded, we no longer need to use technology to control, monitor, and steal it from anyone and can

fulfill original goals of accessibility and equalization – not only serving ourselves, but our communities and environments as well.

Alternative Labor: Care, Commons, and Domestic Labor

Where capitalism demands labor in exchange for survival, other systems have centered mutual care, reciprocity, and use-value over profit. David Graeber's anthropological work shows how gift economies (systems without formal currency) have long thrived through trust, community, and collective memory. Silvia Federici's analysis of reproductive labor exposes how capitalism exploits unquantified work (caregiving, cleaning, emotional support) to keep formal economies afloat. (Federici, 2012)

Karl Marx offers a powerful critique in his theory of use-value versus exchange value. He draws a distinction between what is useful and what is profitable. Under capitalism, something is only valuable if it can be sold, regardless of whether it actually improves life. A loaf of bread feeds you, but a designer handbag does not. However, the handbag carries a higher exchange value. Similarly, a grandmother's wisdom, a nurse's care, or a community's ability to mourn together all have immense use-value, but they're rendered invisible and steadily marginalized, denied, and replaced by products because they don't circulate on the market. (Marx, 1867/1976) Not only does this disempower workers economically, but it leads to a dissatisfaction, what Marx called "alienation," in workers' lives. The worker feels purposeless and unsatisfied with their accomplishments and purpose and feels disconnected and unattached to the products they have played a part in creating. As no worker is responsible for the entirety of a product or creation, nor may they even understand how a product works since they are only responsible for a small

piece, they experience a dissonance between themselves and their work and this leads to a dissonance and lack of satisfaction in their lives and, as a result, through all of society. (Marx, 1867/1976) This economic structure doesn't just shape our jobs. It reshapes our sense of worth. When only commodified labor is honored, we are taught to distrust idleness, rest, or joy. Productivity becomes a moral duty. Under these conditions, leisure is not laziness – it's resistance.

Alternative Futures: Dreaming New Worlds

In Frederic Jameson's 2003 essay *Future City*, he says "Someone once said that it is easier to imagine the end of the world than to imagine the end of capitalism," and he is frequently cited as the originator of this quote. In Ursula K. Le Guin's acceptance speech for the National Book Foundation's Medal for Distinguished Contribution to American Letters in 2014, she said "we live in capitalism, its power seems inescapable – but then, so did the divine right of kings. Any human power can be resisted and changed by human beings. Resistance and change often begin in art. Very often in our art, the art of words." Her fiction, similar to Octavia Butler's, doesn't strictly imagine dystopias. It also imagines solutions and alternatives. Butler's *Parable* series envisions how communities survive collapse not because of AI or technology, but through adaptability, care, and belief in change. These fictional futures are not simple escapism but also inspiration and ideas for what could be. They challenge us to reimagine and remember what technology could be meant for. Was it always meant to be a tool for corporate extraction? Or was it something we wanted to meet real human and planetary needs?

In *The Dispossessed*, Le Guin introduces us to two neighboring societies, one based on the planet Urras and the other on its moon, Anarres. (Le Guin, 1974) Urras has two major societies - one is capitalist and patriarchal and the other is authoritarian, while Anarres has an anarchist cooperative society. While Anarres' anarchist society is not perfect and grapples with scarcity, bureaucracy, and social pressures, at its foundation, it is built on mutual aid, shared labor, and the belief that no one owns anything, including time. Time itself is reorganized, and work is seasonal, voluntary, and coordinated through communal discussion, not imposed by wage labor nor profit margins. In this imagined anarchist world, Le Guin offers a model for how technology, science, and community might exist without exploitation.

Octavia Butler's *Parable of the Sower* and *Parable of the Talents* present a very different vision, one of collapse. (Butler, 1993/1998) Starting in 2024 in the United States of America, *Parable of the Sower* begins when society is unraveling from climate change, economic inequality, and social decay. Butler's world feels prophetic. Amid the wreckage, her protagonist, Lauren, nurtures a new belief system, one that centers on adaptability, interdependence, and long-term care – that God is Change and that God can be shaped by believers. The communities that emerge do not survive through technological dominance, but through collective knowledge, practical skills, emotional intelligence and a refusal to give up on one another. They garden, teach, defend, and change. Because they change, they endure. While Butler's characters face hardship, struggle, and sorrow through both novels, their stories are also about revisioning the present and the future and making

choices to change them together, through by developing relationships and creating cooperative groups.

Both of these authors, and others, remind us that we don't need permission to reimagine the future. Just as techno-capitalists are trying to bring to life dystopias most people see as cautionary tales, we too can imagine and build different worlds. Worlds where time belongs to people, not profit. Where labor sustains life instead of extracting it. Where technology is reclaimed as a tool of care, not control. That future can begin now.

What Future Are We Building?

Time was never neutral. It has been counted, sliced, and sold – turned into wages, punishments, deadlines, and quotas. Similarly, AI is not a neutral tool. Rather than liberating us from the clock, it tightens the grip. Marketed as time-saving, it commodifies even our attention and imagination, asking us to work faster, say less, and feel nothing. It is a technology shaped by the values of those who build and deploy it. Despite being marketed to reclaim our time and enhance productivity, artificial intelligence often does the opposite. It extends the logic of extractive capitalism, compressing our labor, erasing rest, and monetizing even our thoughts. It continues the promises of the industrial age: faster, better, more efficient. But instead, it entrenches the systems it claims to transcend.

But other models of time exist. So do other models of labor, value, and life. We are not trapped. The future is not fixed. We can and must imagine technologies that restore time instead of stealing it. If we listen to our ancestors, to the Earth, to stories that spiral instead of march, we may find our way out of this trap. This trajectory was never inevitable.

Many cultures understand time not as a linear construct, but as a spiral, a season, or a rhythm. Labor, too, can be reimagined as something no longer exploitative, but instead as care, mutuality, or contribution to the commons. If we approach AI critically as a symptom instead of salvation, we can begin to ask better questions. Instead of “how do we keep up?” we could ask “what kind of future do we want to live in?” Instead of “how can AI replace us?” we can ask “how can we reclaim the time it steals?” We can ask “how can we care for the next seven generations?”

The answers to those questions may not lie in the next model or platform, but in the stories we tell, the values we center, and the communities we build. This is not a technological problem – it's a human one. When we remember the alternatives we had, that we could still have, and the people who have made changes before us, it's so much easier to dream and create the new futures we want to step into. Ursula K Le Guin's *The Dispossessed* is an example. And Octavia Butler's *Parable* series is an example.

Everything was SUPPOSED to get easier. That it didn't was a choice, and choices are political. But we, the politic, can make a different choice.

BIBLIOGRAPHY

Alimardani, A. (2024). Evaluating generative AI in legal education. *Australian Journal of Law and Technology*.

Anthropic. (2024). Agentic misalignment: An emerging risk.
<https://www.anthropic.com/research/agentic-misalignment>

Arxiv authors. (2025). Prompt adaptation for aligning agents with human preferences. *arXiv*. <https://arxiv.org/abs/2507.07935>

Baer, J. (2014). *Youtility: Why smart marketing is about help not hype*. Portfolio.

Blix, H., & Glimmer, I. (2025). *Why we fear AI: On the interpretation of nightmares*. Common Notions.

Bryan, K. (2025, July 2). Meta's AI climate tool raised false hope of CO2 removal, scientists say. *Financial Times*. <https://www.ft.com/content/69220fea-9c9a-4f55-9118-65d2660c85d4>

Butler, O. E. (1993/1998). *Parable of the Sower & Parable of the Talents*. Grand Central Publishing.

CIO Dive. (2023, November 15). How accessibility drives innovation. *CIO Dive*.
<https://www.ciodive.com/spons/how-accessibility-drives-innovation/696299/>

Cornell Lab of Ornithology. (n.d.). *Merlin Bird ID*. <https://merlin.allaboutbirds.org/>

Cowan, R. S. (1983). *More work for mother: The ironies of household technology from the open hearth to the microwave*. Basic Books.

Crawford, K. (2021). *Atlas of AI: Power, politics, and the planetary costs of artificial intelligence*. Yale University Press.

DAIR Events. (2022, December 20). *DAIR's first anniversary - Adrienne Williams, Surveillance nation: The real cost of Amazon's customer obsession* [Video]. PeerTube.
<https://peertube.dair-institute.org/w/oAWnan9PXTvS3CyutatQcg>

Dellinger, A. J. (2025, July 14). Meta is building massive gigawatt data centers as it pours billions into AI. *Gizmodo*. <https://gizmodo.com/meta-is-building-massive-gigawatt-data-centers-as-it-pours-billions-into-ai-2000628918>

Deloria, V. Jr. (2003). *God is red: A Native view of religion* (3rd ed.). Fulcrum Publishing.

Doctorow, C. (2025). AI can't do your job. *Pluralistic Daily*.

<https://pluralistic.net/2025/03/18/asbestos-in-the-walls/#government-by-spicy-autocomplete>

Dollinger, A. J. (2025, June 26). Salesforce CEO claims half of the company's work is now done by AI. Gizmodo. <https://gizmodo.com/salesforce-ceo-claims-half-of-the-companys-work-is-now-done-by-ai-2000620730>

Fauna & Flora International. (n.d.). What is conservation technology? How tech solutions can protect the world's wildlife. <https://www.fauna-flora.org/explained/what-is-conservation-technology-how-tech-solutions-can-protect-the-worlds-wildlife/>

Fortunato, L., & Galassi, M. (2021). The case for free and open source software in research and scholarship. *Philosophical Transactions of the Royal Society A*, 379(2197), 20200079. <https://doi.org/10.1098/rsta.2020.0079>

Ghosh, A. (2021). *The nutmeg's curse: Parables for a planet in crisis*. University of Chicago Press.

Graeber, D. (2011). *Debt: The first 5,000 years*. Melville House.

Green Technology Info. (2023, November 14). What is green technology? A beginner's guide to sustainability. <https://www.greentechnologyinfo.com/what-is-green-technology-a-beginners-guide-to-sustainability/>

IBM. (2025, May 6). IBM study: CEOs double down on AI while navigating enterprise hurdles. IBM Newsroom. <https://newsroom.ibm.com/2025-05-06-ibm-study-ceos-double-down-on-ai-while-navigating-enterprise-hurdles>

iNaturalist. (n.d.). *iNaturalist*. <https://www.inaturalist.org/>

iNaturalist. (2025, June 10). iNaturalist educator's guide. <https://help.inaturalist.org/en/support/solutions/articles/151000170805-inaturalist-educator-s-guide>

The Indigenous Foundation. (2021, June 30). *Seven generations principle: Healing the past & shaping the future*. <https://www.theindigenousfoundation.org/articles/seven-generations-principle-healing-the-past-amp-shaping-the-future>

Ivanova, I. (2025, May 9). As Klarna flips from AI-first to hiring people again, a new landmark survey reveals most AI projects fail to deliver. *Fortune*.

<https://fortune.com/2025/05/09/klarna-ai-humans-return-on-investment/>

Koebler, J. (2024, April 17). *Bryan Johnson is now removing all his blood*. *Futurism*.

<https://futurism.com/neoscope/bryan-johnson-removing-all-blood>

Jacobs, S. (2025, June 3). Builder.ai collapses after revelation that its "AI" was hundreds of engineers. *TechSpot*. <https://www.techspot.com/news/108173-builderai-collapses-after-revelation-ai-hundreds-engineers.html>

Kimmerer, R. W. (2013). *Braiding sweetgrass: Indigenous wisdom, scientific knowledge, and the teachings of plants*. Milkweed Editions.

Le Guin, U. K. (1969). *The left hand of darkness*. Ace Books.

Le Guin, U. K. (1974). *The dispossessed*. Harper & Row.

Le Guin, U. K. (2014, November 19). [Acceptance speech for Medal for Distinguished Contribution to American Letters]. *National Book Foundation*.

<https://www.ursulaklequin.com/nbf-medal>

MacAskill, E. (2024, July 2). *Two warring visions of artificial intelligence*. *Prospect Magazine*.

<https://www.prospectmagazine.co.uk/ideas/technology/64491/two-warring-visions-of-artificial-intelligence-tescreal>

Marx, K. (1867/1887). *Capital: A critique of political economy, Volume I, Book One: The process of production of capital* (S. Moore & E. Aveling, Trans.; F. Engels, Ed.). Progress Publishers. <https://www.marxists.org/archive/marx/works/1867-c1/>

Marx, K. (1978). Estranged labor. In R. C. Tucker (Ed.), *The Marx-Engels reader* (2nd ed., pp. 70–81). W. W. Norton.

Meta AI. (2025). Detection of CO2 materials by AI [Archived]. <https://archive.is/8sYNv>

Metz, C., & Weise, K. (2025, May 5). A.I. is getting more powerful, but its hallucinations are getting worse. *The New York Times*. <https://www.nytimes.com/2025/05/05/technology/ai-hallucinations-chatgpt-google.html>

Microsoft. (2025, June). Internal memo: Using AI is no longer optional. *Business Insider*. <https://www.businessinsider.com/microsoft-internal-memo-using-ai-no-longer-optional-github-copilot-2025-6>

Mohammed, Moyasser Y and Mirosław J Skibniewski. "The Role of generative AI in Managing Industry Projects: Transforming Industry 4.0 Into Industry 5.0 Driven Economy." *Law and Business*, vol 3, no 1, January 2023, pp 27 – 41, doi.org/10.2478/l...

More Perfect Union [@moreperfectunion]. (2025, July 23). *Demand for data centers is booming as ChatGPT and other AI tools become integrated into our daily life. The Trump administration is investing over \$1 trillion into new data...* [Video]. TikTok. <https://www.tiktok.com/@moreperfectunion/video/7530341085252504862>

Salesforce CEO. (2025). Half of the company's work is now done by AI. *Gizmodo*. <https://gizmodo.com/salesforce-ceo-claims-half-of-the-companys-work-is-now-done-by-ai-2000620730>

Scheiber, N. (2025, May 25). At Amazon, some coders say their jobs have begun to resemble warehouse work. *The New York Times*. <https://www.nytimes.com/2025/05/25/business/amazon-ai-coders.html>

Specht, M. (2019, October 3). Natural gas power plants are not clean. *Union of Concerned Scientists Blog*. <https://blog.ucs.org/mark-specht/natural-gas-power-plants-are-not-clean/>

Stewart, A. (2025, June 27). Microsoft pushes staff to use internal AI tools more, and may consider this in reviews. 'Using AI is no longer optional.' *Business Insider*. <https://www.businessinsider.com/microsoft-internal-memo-using-ai-no-longer-optional-github-copilot-2025-6>

Thompson, E. P. (1993). Time, work-discipline, and industrial capitalism. In *Customs in common: Studies in traditional popular culture* (pp. 352–403). The New Press.

Torres, É. P. (2021, July 30). *The dangerous ideas of “longtermism” and “existential risk”*. *Current Affairs*. <https://www.currentaffairs.org/news/2021/07/the-dangerous-ideas-of-longtermism-and-existential-risk>

University of Washington. (n.d.). *Foldit*. <https://fold.it/science>

Williams, A., Miceli, M., & Gebru, T. (2022). The exploited labor behind artificial intelligence: Supporting transnational worker organizing should be at the center of the fight for ethical AI. *Noema*. <https://www.noemamag.com/the-exploited-labor-behind-artificial-intelligence/>